SENG 201
Software Engineering Foundations

[Week 1, Tuesday]



Labs and tutorials start today

QuizServer: Self-enrol with key `seng201`

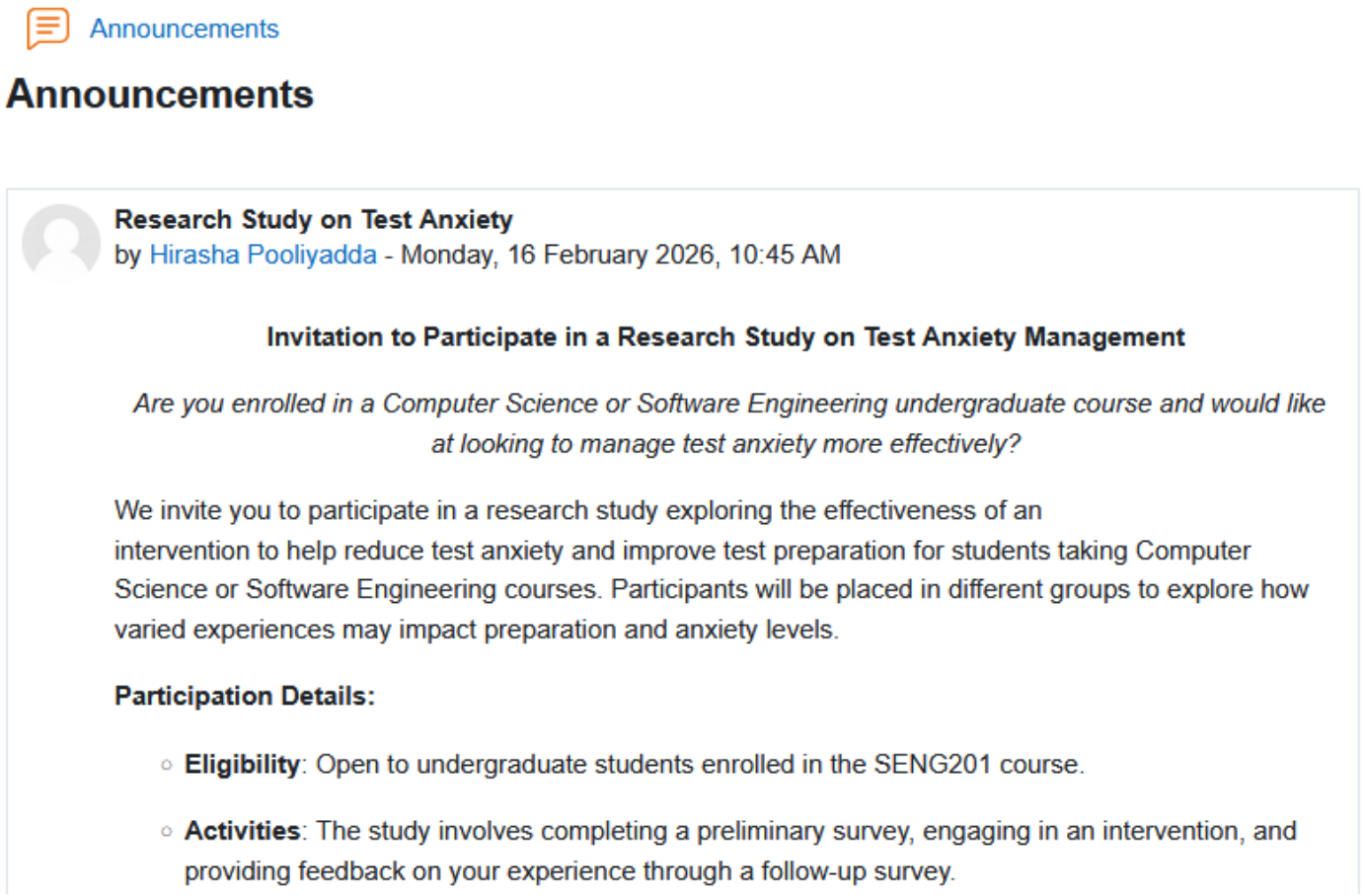
Bonus: 5 marks on the Lab Test

Conditions:

- Complete all labs and tutorials (incl. Term 2)
- 90% score to count as complete
- 2.5 weeks to complete (closes on Friday)
- Unlimited attempts with no penalty

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Consider enrolling in Hirasha's study



Announcements

Announcements

Research Study on Test Anxiety
by Hirasha Pooliyadda - Monday, 16 February 2026, 10:45 AM

Invitation to Participate in a Research Study on Test Anxiety Management

Are you enrolled in a Computer Science or Software Engineering undergraduate course and would like at looking to manage test anxiety more effectively?

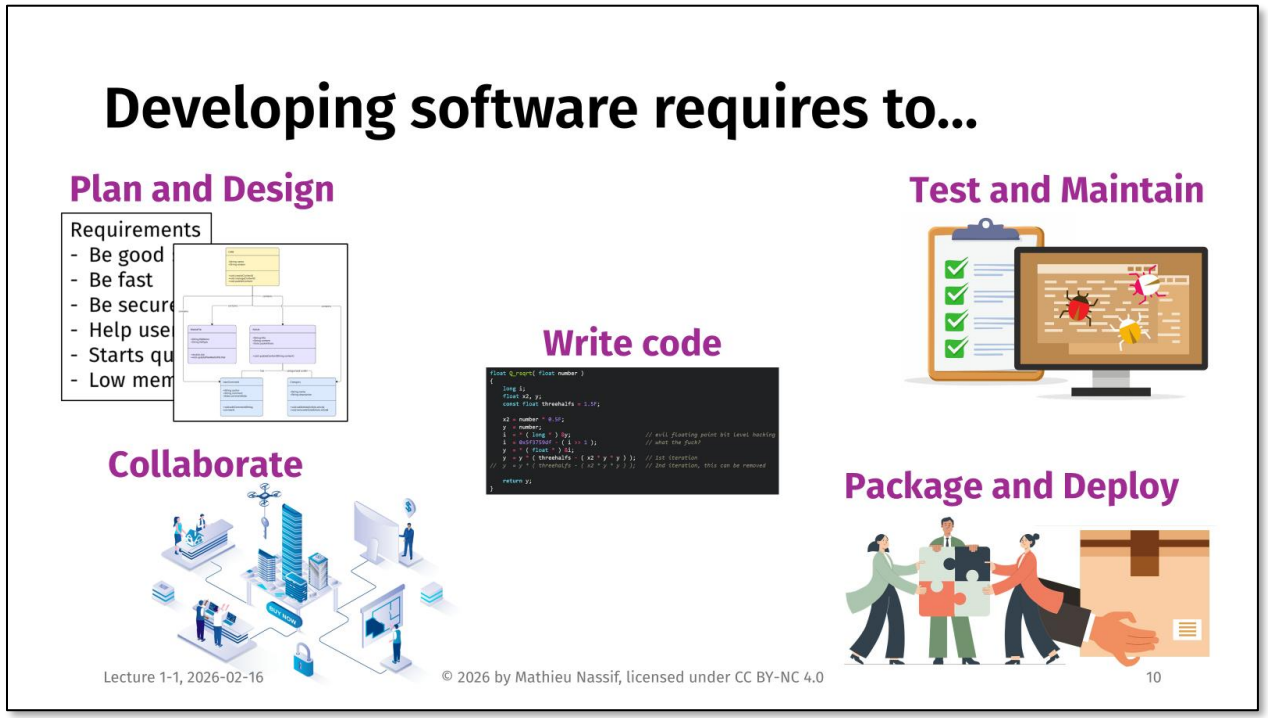
We invite you to participate in a research study exploring the effectiveness of an intervention to help reduce test anxiety and improve test preparation for students taking Computer Science or Software Engineering courses. Participants will be placed in different groups to explore how varied experiences may impact preparation and anxiety levels.

Participation Details:

- **Eligibility**: Open to undergraduate students enrolled in the SENG201 course.
- **Activities**: The study involves completing a preliminary survey, engaging in an intervention, and providing feedback on your experience through a follow-up survey.

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Recall from yesterday...



How and when do we plan and perform these activities?

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Software development processes



→ Processes and activities

→ Software development lifecyle (SDLC)

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Processes, activities, tasks

How and when do we plan and perform each development activities?

~~ad hoc~~

Process *"Set of interrelated or interacting **activities** that transform inputs into outputs"*

Activity *"Set of cohesive **tasks** of a process"*

Task *"Required, recommended, or permissible **action**, intended to contribute to the achievement of one or more outcomes of a process"*

AS/NZS ISO/IEC/IEEE 15288:2023

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System requirements definition process

1. Prepare for system requirements definition. — Activity
   a. Define the functional boundary of the system. — Task
   b. Define the system requirements definition strategy. — Task
   c. …
2. Define system requirements. — Activity
   1. Define each function that the system is required to perform.
   2. Define necessary implementation constraints.
   3. …
3. Analyse system requirements.
4. …

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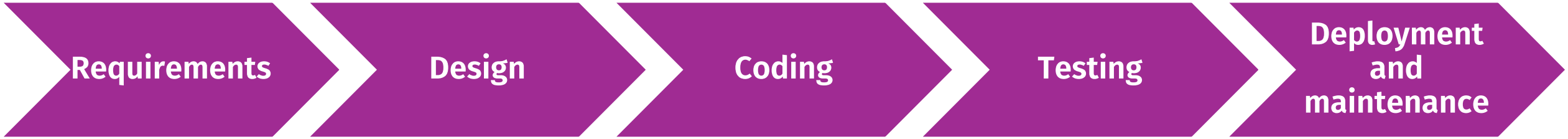
Software Development Lifecycle (SDLC)

SDLC model  **High-level process**

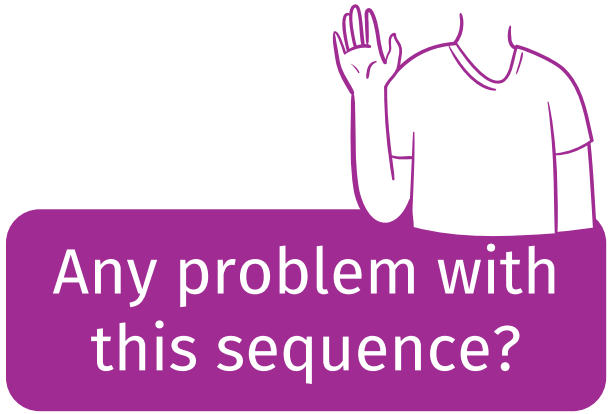
- affects development culture, philosophies, tooling
- start of project to end of support

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A list of SDLC activities*



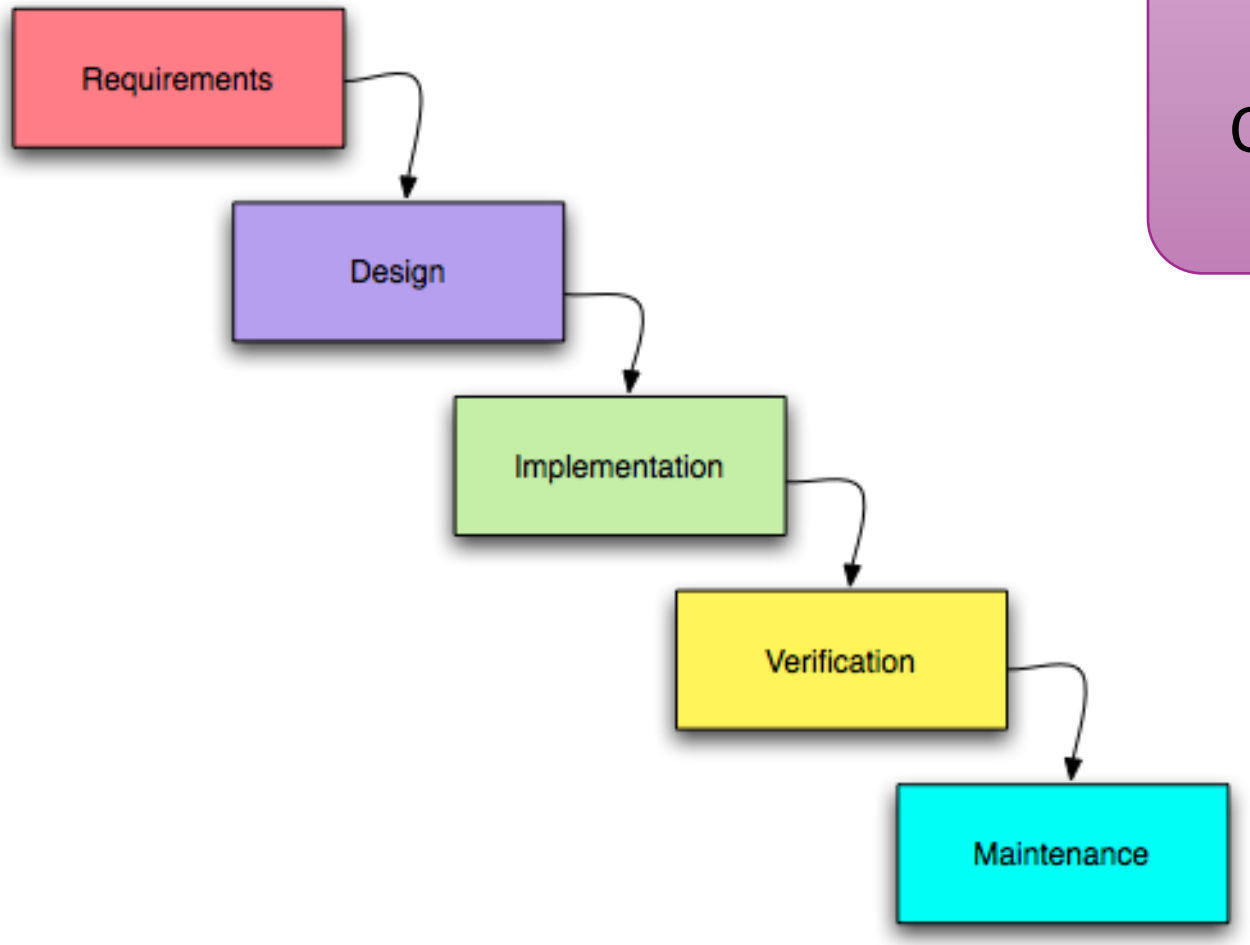
*Different terminologies exist



We never get it right the first time

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Example 1: Waterfall

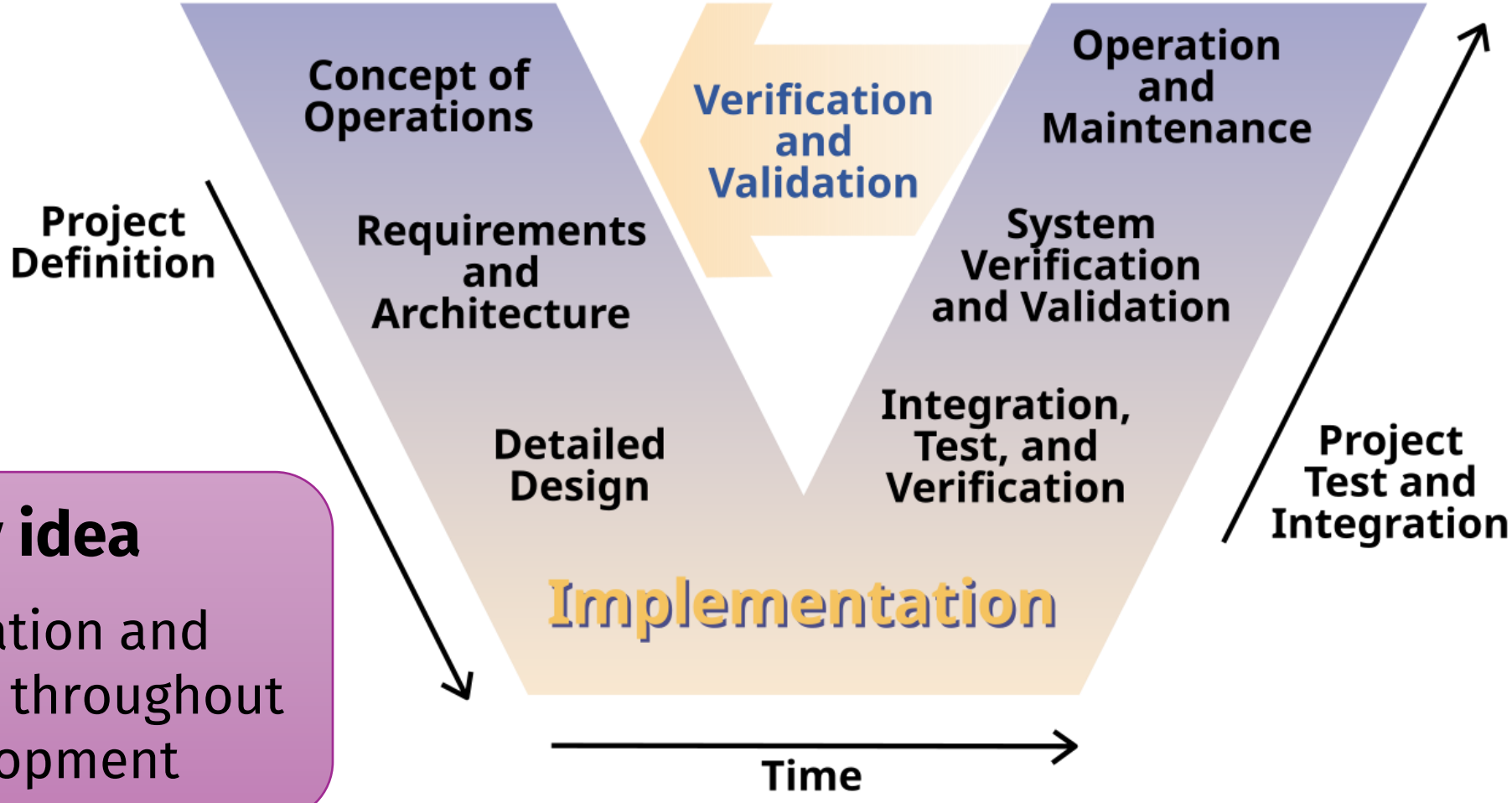


Key idea

Clear separation of phases/activities

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Example 2: V-Model

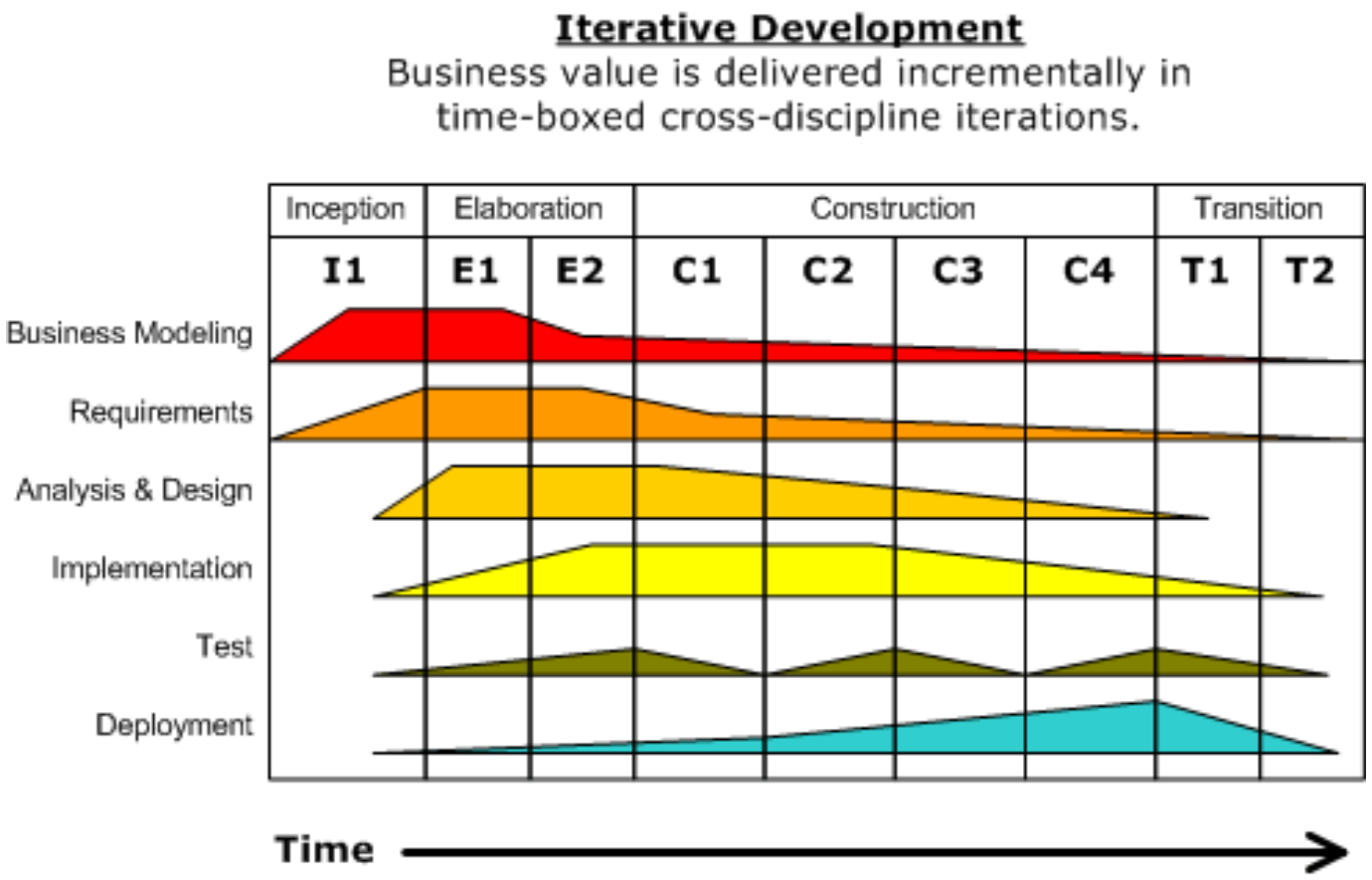


Key idea

Verification and validation throughout development

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Example 3: Rational Unified Process



Key idea

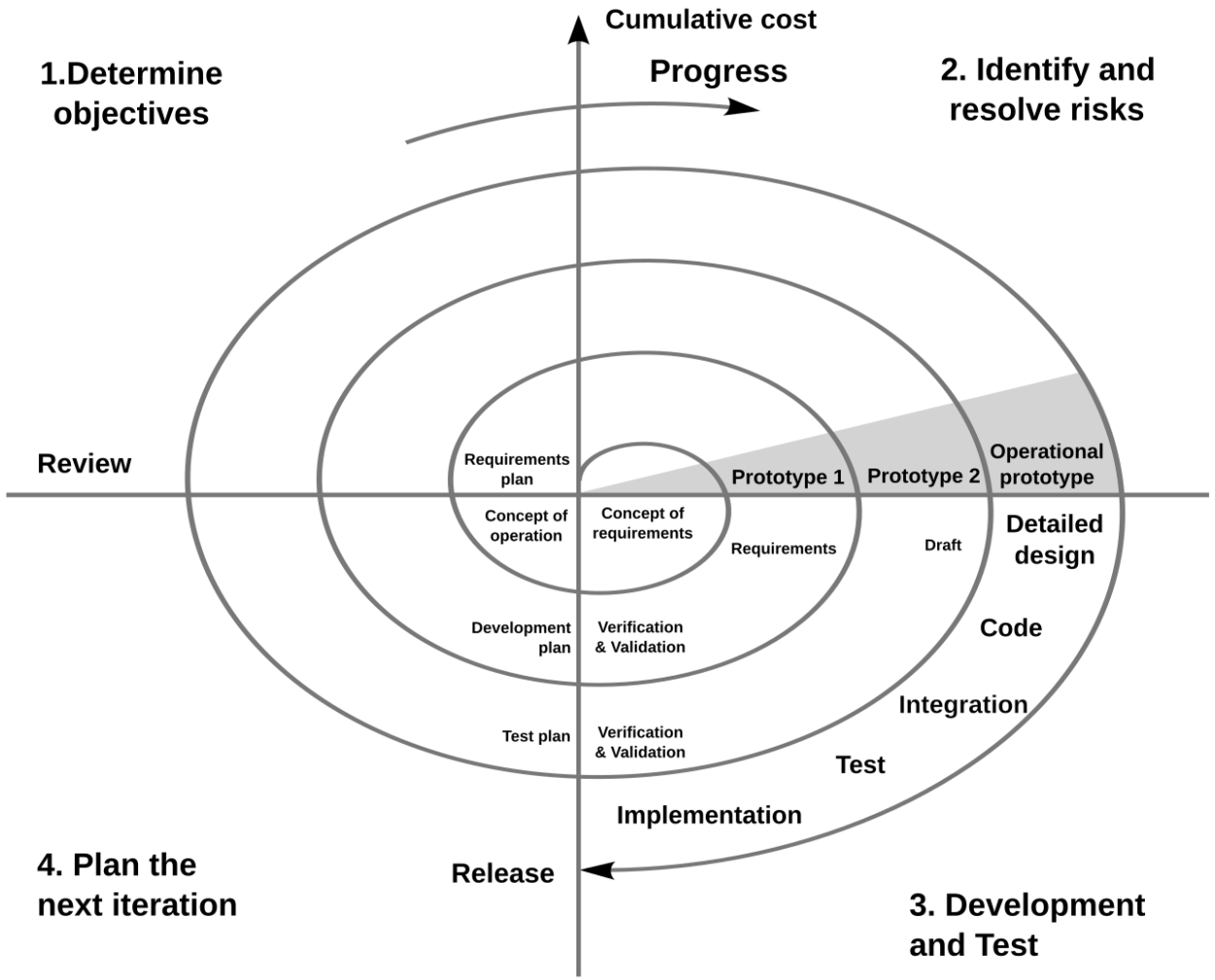
Iterative, component-based development with clear objectives for each phase

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Example 4: Spiral

Key idea

Risk-driven, iterative process

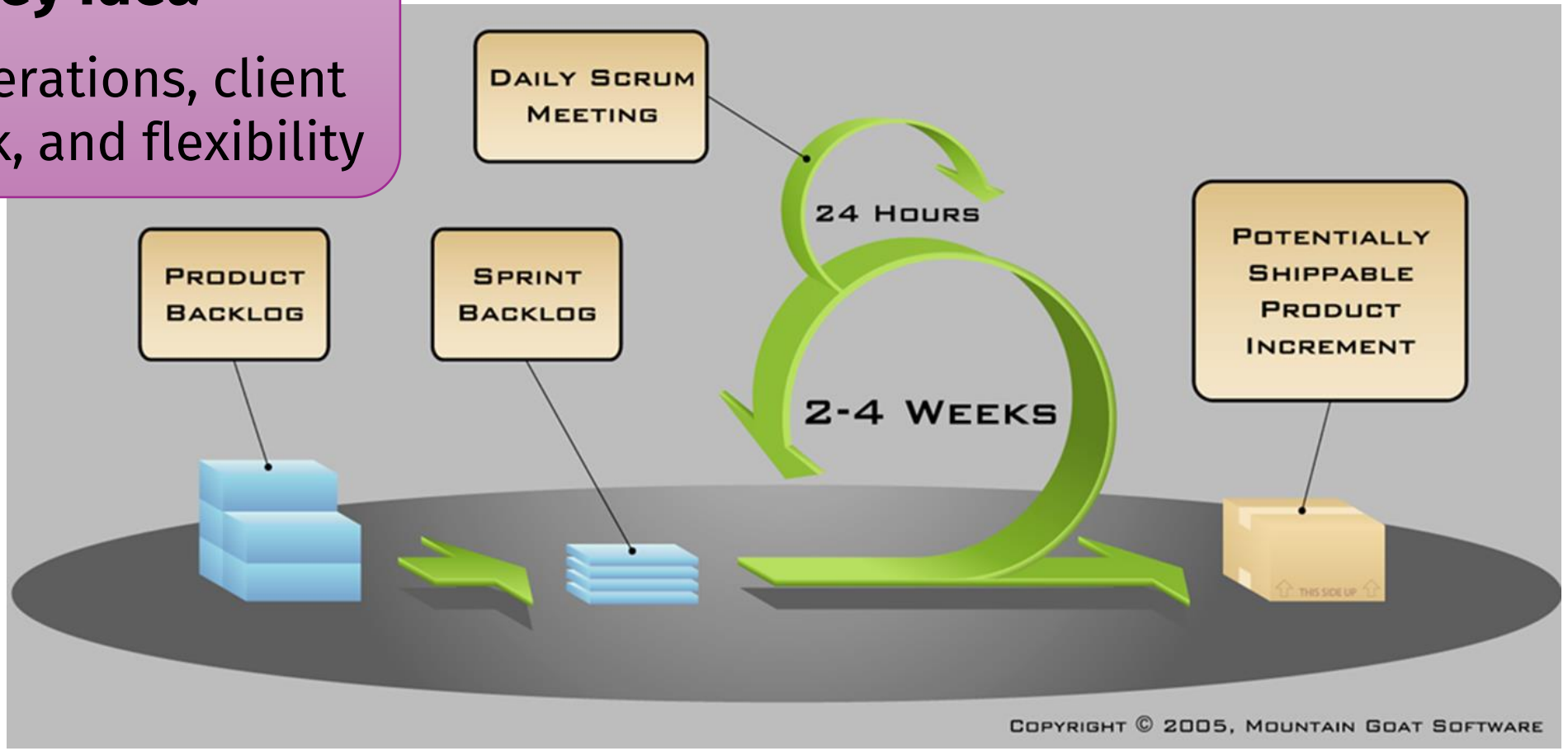


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Example 5: Scrum (agile process)

Key idea

Short iterations, client feedback, and flexibility

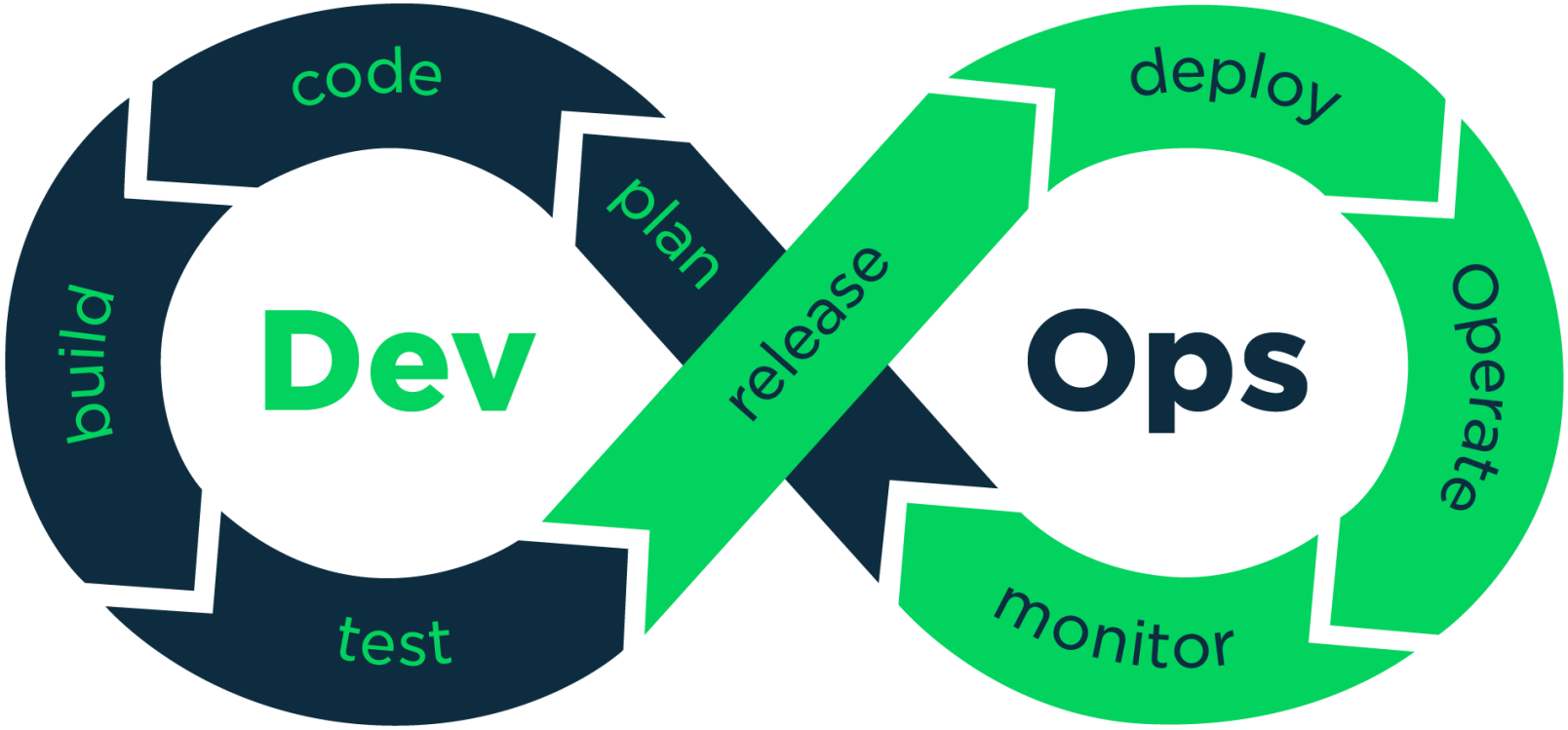


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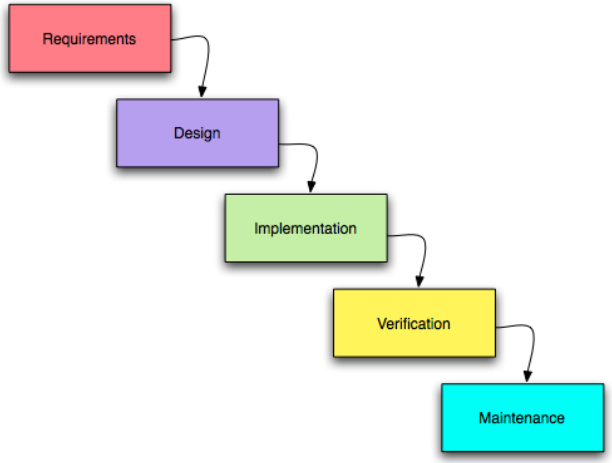
Example 6: DevOps

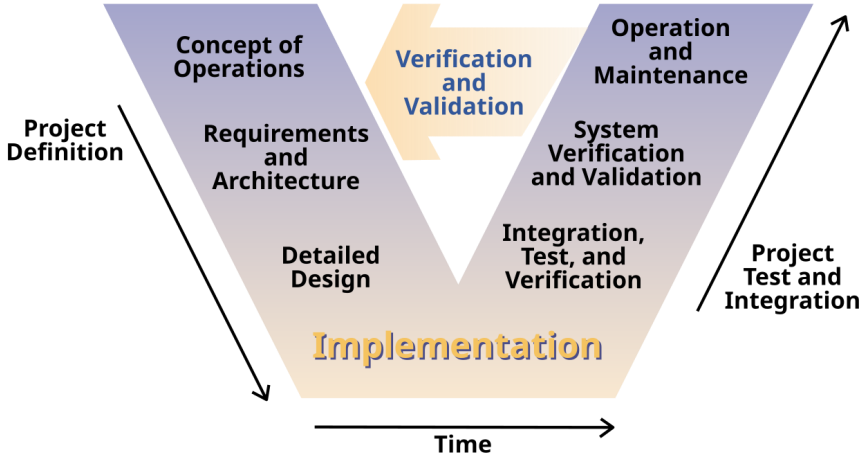
Key idea

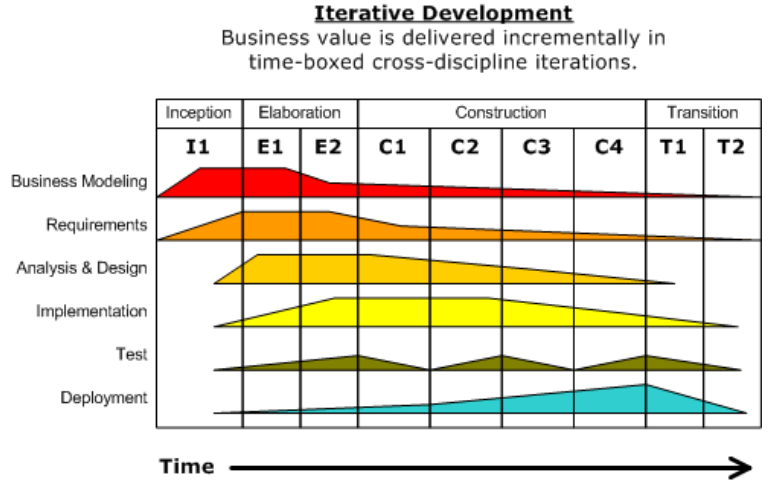
Automated, small, frequent deployments, integrate with development

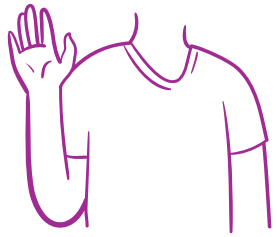


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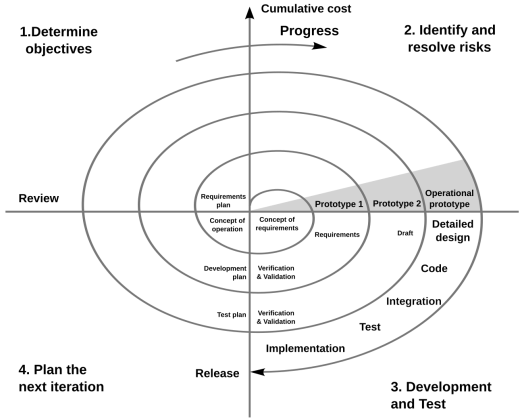


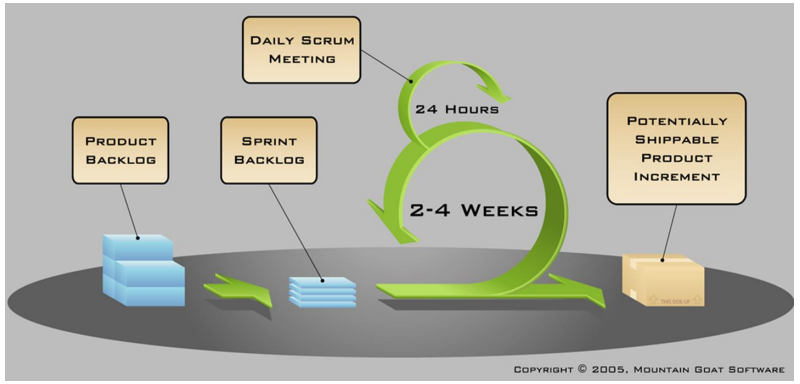


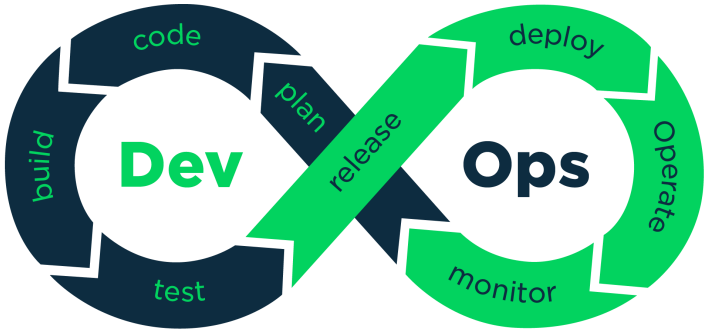




Which process(es) should you use?

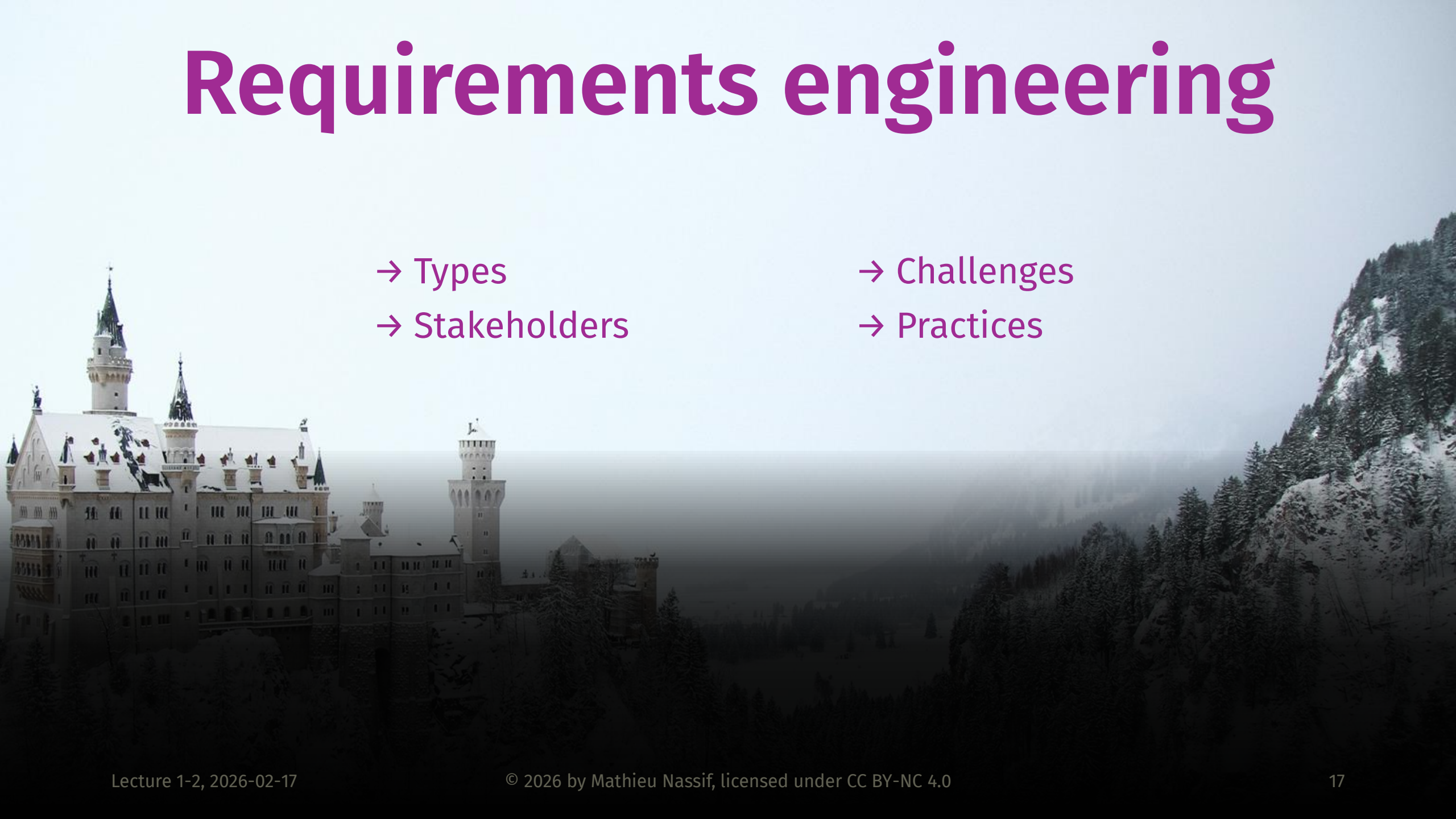






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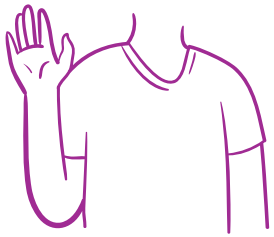
Requirements engineering



→ Types
→ Stakeholders

→ Challenges
→ Practices

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What's a requirement?
What are examples of requirements?

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Requirements

Definition: An explicit, clear, objective condition that must be satisfied for the project's output to be acceptable

Functional

- Input
- Output
- Data handling
- Use cases

- Clients must be able to search blogs
- Account management scenario
- Invalid inputs and error handling

Quality (non-functional)

- Dev. Process
- Product/Technical
- External/Business

- Delivery
- Standards
- Reusability
- Security
- Scalability
- Time-to-market
- Legal constraints
- Budget

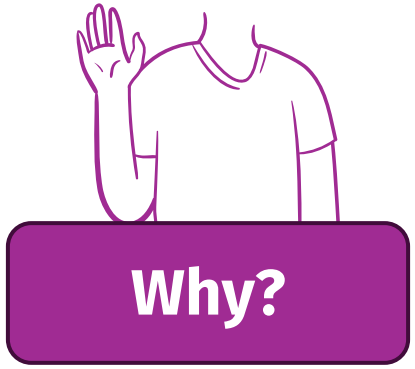
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Specifying requirements is hard!

Non technical clients
Domain experts

Implicit expectations
Forgotten requirements

Conflicting constraints
(time, budget, quality)



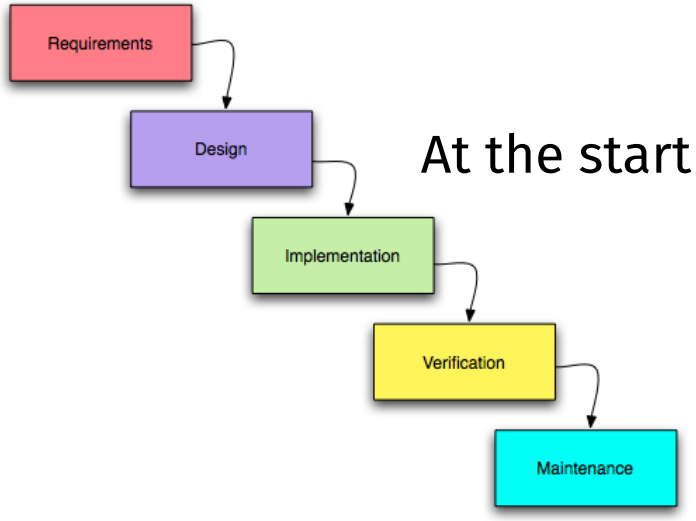
Vague, ambiguous needs

Uneven priorities

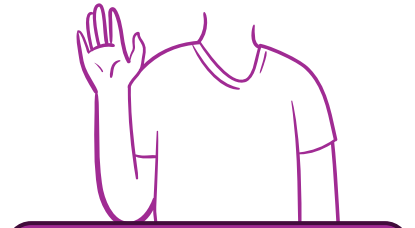
Changing environment

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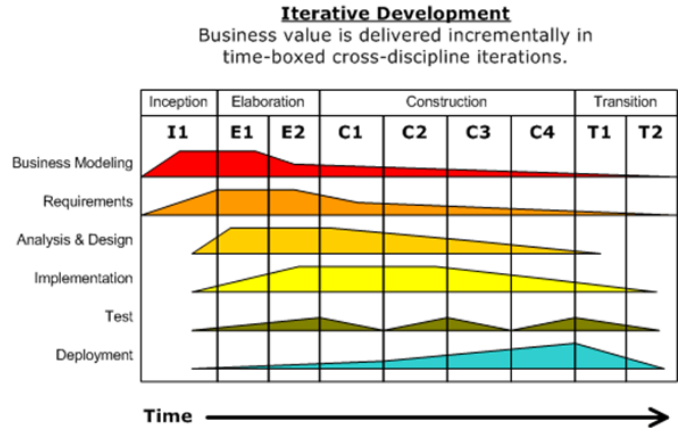
Specifying requirements is hard!



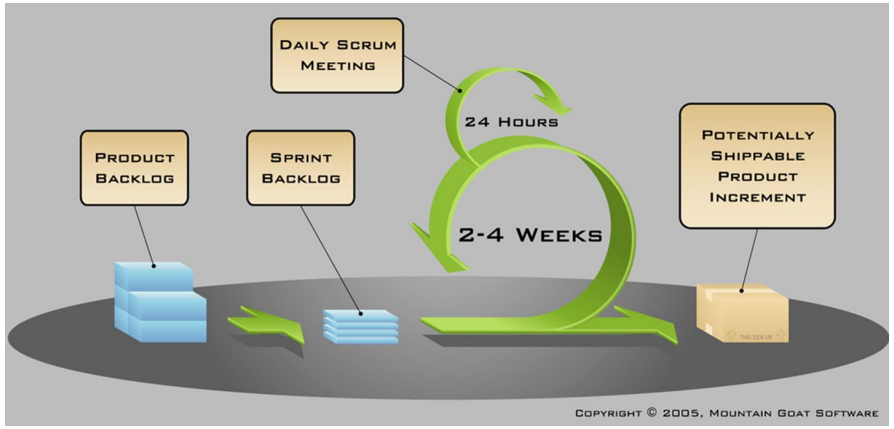
At the start



When?



Multiple times, iteratively/incrementally



Continuously

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How to specify requirements?

Stakeholder analysis: anyone affected by the project

➢ clients, developers, …



Requirements elicitation

➢ meetings, …



Requirements documentation

➢ formal specifications, user stories, UML diagrams

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How to specify requirements?

Requirements analysis

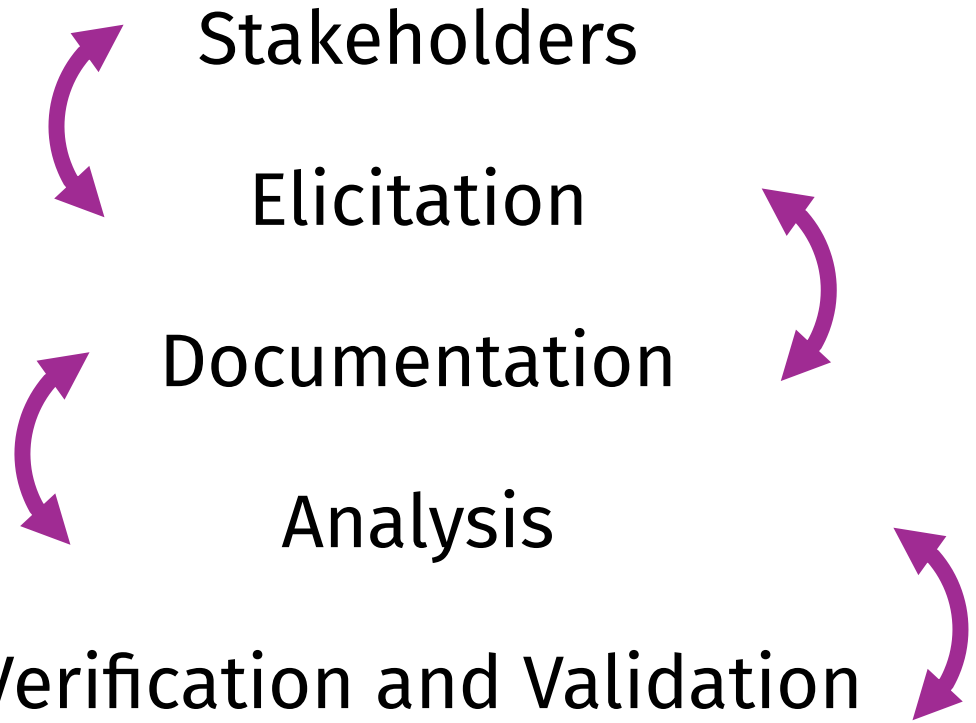
- Prioritize
- Estimate effort
- Identify dependencies

Requirements verification and validation

- “Feasible”: Realistic, unambiguous, non-conflicting, …
- “Right”: what the client truly wants, what the users truly need

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How to specify requirements?



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Let's try...



Ride sharing app

Stakeholders

- Drivers
- Passengers
- Other drivers
- Taxi companies
- Uber Technologies, Inc.
- Uber employees
- Payment services
- Local regulators

Functional requirements

- Order a ride
- Accept a ride
- Cancel a ride
- Be notified of change events
- Sign up, log in
- Add payment information
- Delete account
- Gather data for fraud analysis

Quality requirements

- Data protection
- Payment security
- Transaction traceability
- Scalability
- Maximum latency
- Reliability

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Study on Test Anxiety in CS and SE Education

UNIVERSITY OF CANTERBURY
Te Whare Wānanga o Waitaha
CHRISTCHURCH NEW ZEALAND

- **Objectives**
  - **Gain insights** into undergraduate **current experiences with test anxiety in programming tests** and its impact.
  - **Explore the effectiveness of** the proposed **intervention in reducing test anxiety** among CS and SE students.
  - Assess how the intervention influence mental well-being and academic performance.

- **Research team**
  - Hirasha Pooliyadda (PhD student)
  - Valerie Sotardi (School of Curriculum, Pedagogy and Assessment)
  - Miguel Morales

Slides courtesy of Hirasha Pooliyadda

RECAP

→ Processes are structured sets of activities

→ Multiple SDLC models and processes

→ Requirements can be changing, ambiguous, varied

→ Many stakeholders must be considered



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